

Universal Properties from Quantum Many-Body Dynamics

Master's Thesis Defence

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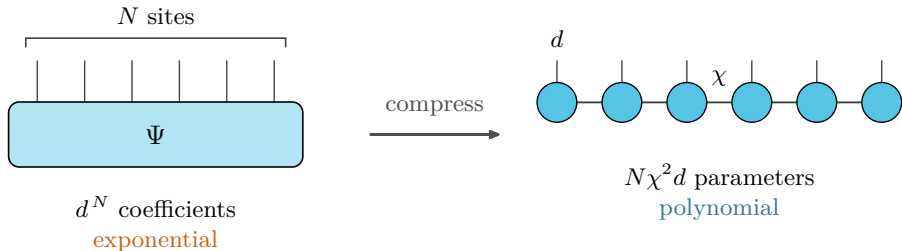
**Barcelona
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**Master in Quantum
Science and Technology
Barcelona**

Simulating Quantum Many-Body Systems



tensor networks organise a state by its entanglement

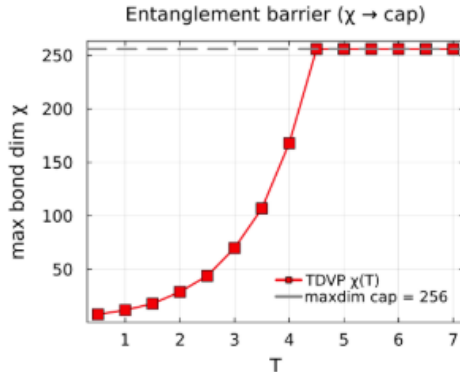
equilibrium in 1D is solved: MPS + DMRG (area law)

What About Dynamics? The Entanglement Barrier

$$\text{quench} \Rightarrow S_x(t) \sim t \Rightarrow \chi \sim e^{ct}$$

TEBD / TDVP: excellent at short times
— then the wall

**this work: time evolution that
avoids the barrier,
for quenches to a critical point in 1D**



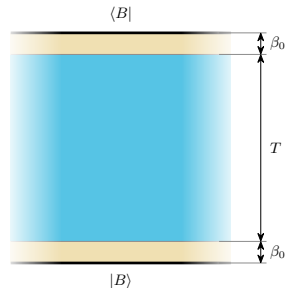
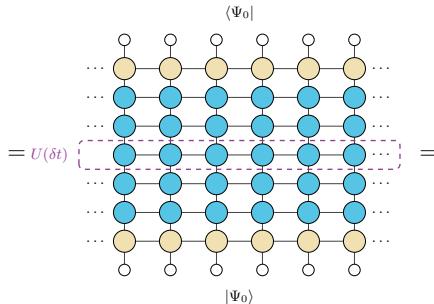
Outline

1. The Loschmidt echo and the CFT mapping
2. Transverse contraction and the CFT predictions
3. The model and its equilibrium properties
4. Numerical methods
5. Results

The Loschmidt Echo

$$\mathcal{L}(\tau) = \langle \Psi_0 | e^{-\tau H} | \Psi_0 \rangle$$

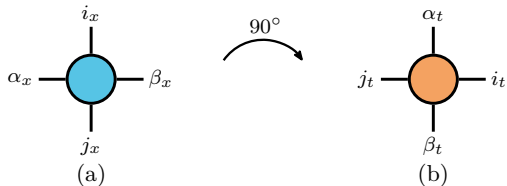
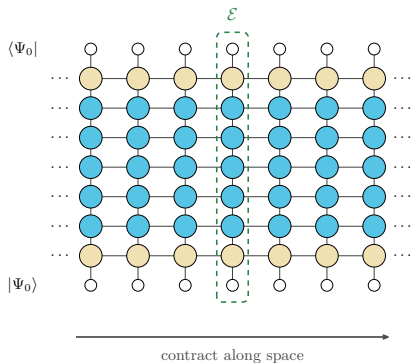
$\tau = \beta_0 + it$



$\tau = \beta_0 + it$: real-time evolution with a cooling regulator β_0

strip width $2\beta = 2v(\beta_0 + iT)$

The Transverse Rotation



virtual space $\rightarrow$ physical *time*
 physical space $\rightarrow$ virtual direction

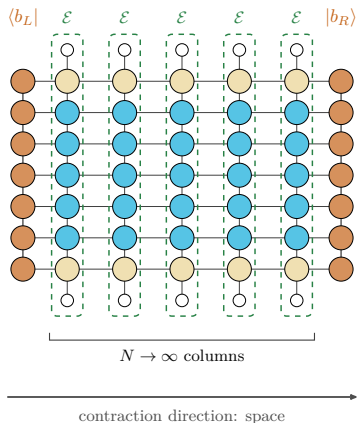
translation invariance $\Rightarrow$ every column is the same operator: the transfer matrix $\mathcal{E}$

Dynamics as an Eigenproblem

$$\mathcal{L}(T) = \langle b_L | \mathcal{E}^N | b_R \rangle \xrightarrow{N \rightarrow \infty} c_0 \mu_0^N$$

$$\mathcal{E} \mathcal{E}^\dagger \neq \mathcal{E}^\dagger \mathcal{E} : \quad \begin{aligned} \langle L | \mathcal{E} &= \mu_0 \langle L | \\ \mathcal{E} | R \rangle &= \mu_0 | R \rangle \end{aligned}$$

everything reduces to the dominant pair and the spectrum of the non-Hermitian transfer matrix $\mathcal{E}$



b_L, b_R : random seeds — the power method converges them to $\langle L |, | R \rangle$

CFT Predictions I — the Leading Eigenvalue

$$\mathcal{E} = e^{-\hat{H}} \quad (a = 1), \quad \hat{H} = \frac{\pi}{\beta} \left(L_0 - \frac{c}{24} \right) \quad \Longrightarrow \quad \mu_i = \exp \left[-\frac{\pi}{\beta} \left(x_i - \frac{c}{24} \right) \right]$$

1. emergent dual unitarity — the moduli lock:

$$\frac{|\mu_i|}{|\mu_0|} = 1 - \frac{b_i}{T^2} \longrightarrow 1$$

2. the dominant phase — the expression we fit:

$$\frac{\text{Im } \lambda_0}{T} = a_0 - \frac{\pi c}{24 v T^2} \Rightarrow c, \quad \lambda_0 = \log(-\mu_0)$$

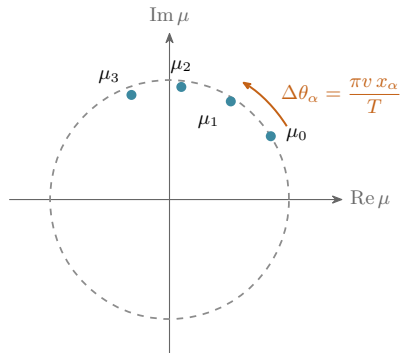
CFT Predictions II — Boundary Operator Content

$$\Delta\theta_\alpha = \arg \mu_\alpha - \arg \mu_0 = \frac{\pi v x_\alpha}{T}$$

each subleading eigenvalue $\longleftrightarrow$
one boundary operator of the CFT

Ising with free boundaries:

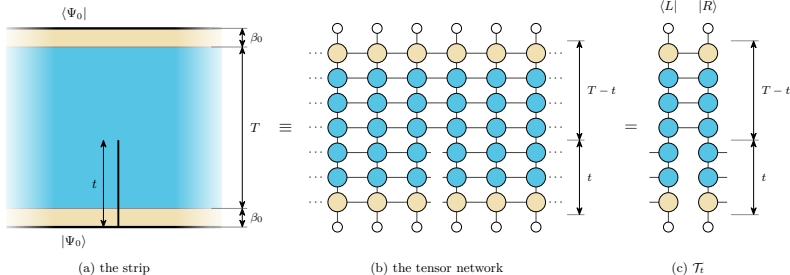
$$x_\alpha \in \left\{ \frac{1}{2}, \frac{3}{2}, 2, \frac{5}{2}, \dots \right\}$$



$$\frac{|\mu_i|}{|\mu_0|} = 1 - \frac{b_i}{T^2} \rightarrow 1$$

at finite T : $|\mu_i| < |\mu_0|$ by $O(\beta_0/T^2)$

Temporal Entropies from the Cut: the RTM



$$\mathcal{T}_t = \frac{\text{Tr}_{>t} |R\rangle \langle L|}{\langle L|R\rangle}$$

$$S_2^{\text{gen}}(t) = -\ln \text{Tr}(\mathcal{T}_t^2)$$

CFT Predictions III — the Temporal Entropy

$$\text{Re } S_2^{\text{gen}}(t) = s_0 + \frac{c}{8} \log \left[\frac{2T}{\pi} \sin \frac{\pi t}{T} \right] \quad (\text{chord dome})$$

$$\text{Im } S_2^{\text{gen}}(t) = \frac{\pi c}{16} \quad (\text{constant plateau})$$

logarithmic in $T \Rightarrow$ polynomial resources

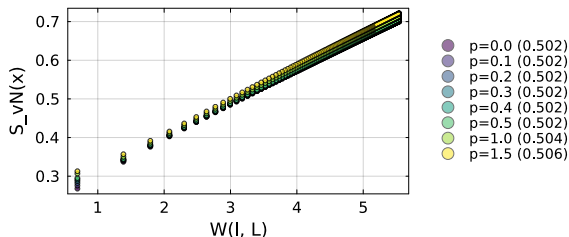
The ANNNI-type Chain Stays in the Ising Class

$$H = - \sum_i \left[\sigma_i^z \sigma_{i+1}^z + \lambda \sigma_i^x + p(\sigma_i^z \sigma_{i+2}^z + \lambda \sigma_i^x \sigma_{i+1}^x) \right], \quad \lambda = 1$$

ground-state entanglement of an open chain:

$$S_{\text{vN}}(l, L) = \frac{c}{6} \log \left[\frac{2L}{\pi} \sin \frac{\pi l}{L} \right] + k$$

**DMRG ground states, $N = 400$,
fit over the middle half of the
chain**



The Sound Velocity

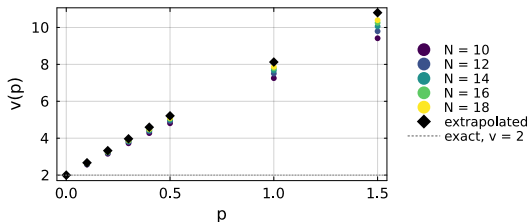
lowest excitation at the smallest
non-zero momentum on a ring:

$$v(N) = \frac{N}{2\pi} [E_1(2\pi/N) - E_1(0)]$$

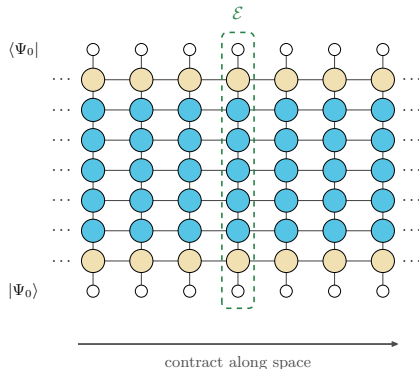
extrapolation, exponent fixed at $p = 0$
where $v = 2$ exactly:

$$v(N) = v_\infty + \frac{C}{N^2}$$

**rings of $N = 10$ – 18 , diagonalised
exactly;
 v_∞ used in every conversion**



A Translation-Invariant Evolution Operator



every column must be the *same* operator

**$U(\delta t)$ must be built from
one repeated bulk tensor W ,
exactly translation invariant**

a second-order construction
sustains $\delta t = 0.1$

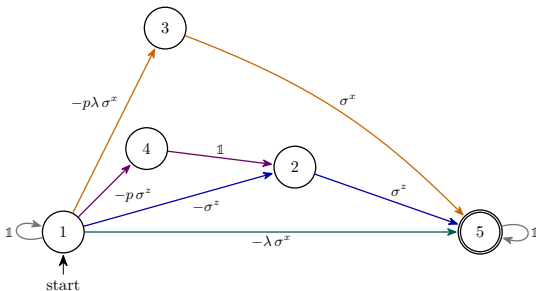
The Hamiltonian as a Finite State Machine

$$W_H = \begin{pmatrix} \mathbb{1} & C & D \\ 0 & A & B \\ 0 & 0 & \mathbb{1} \end{pmatrix}$$

$$= \begin{pmatrix} \mathbb{1} & -\sigma^z & -p\lambda\sigma^x & -p\sigma^z & -\lambda\sigma^x \\ 0 & 0 & 0 & 0 & \sigma^z \\ 0 & 0 & 0 & 0 & \sigma^x \\ 0 & \mathbb{1} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \mathbb{1} \end{pmatrix}$$

$$D_W = 1 + 3 \text{ channels} + 1 = 5$$

idle + {pending σ^z , pending $\sigma^x\sigma^x$, NNN
waiting} + done



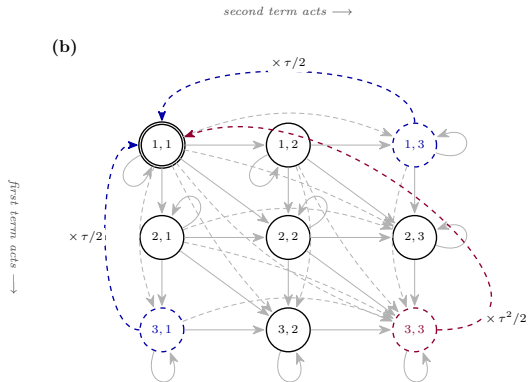
Exponentiating the MPO: the VD2 Construction

$$e^{\tau H} = \mathbb{1} + \tau \sum_x H_x + \frac{\tau^2}{2} \underbrace{\sum_{x,y} H_x H_y}_{\text{incl. overlapping pairs}} + \dots$$

kept through **2nd order**

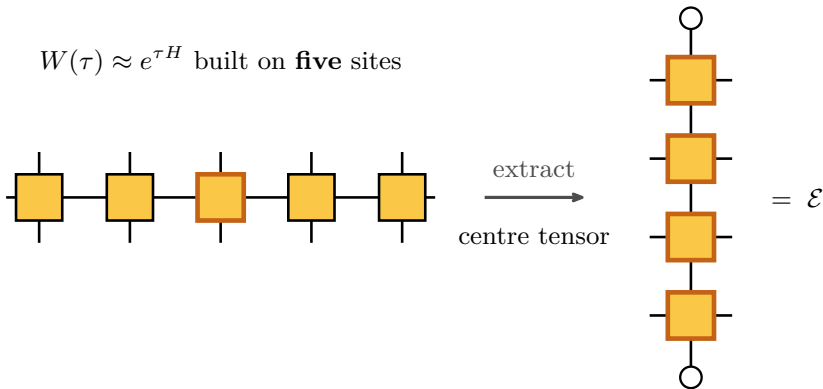
construction of Van Damme et al.,
SciPost Phys. **17**, 135 (2024)

bond = $1 + \chi_H + \chi_H^2 = 13$
measured order 2.00



One Tensor from the Bulk

$W(\tau) \approx e^{\tau H}$ built on **five** sites



the column must be a tensor a bulk site carries, with every interaction present

Extracting the Dominant Pair: the Power Method

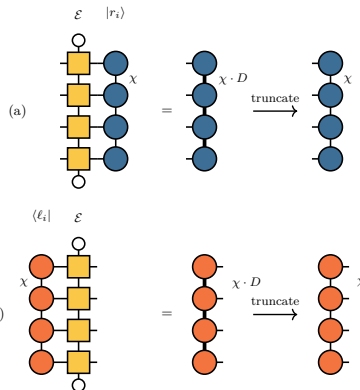
$$\mathcal{E} |r_i\rangle = \mu_i |r_i\rangle, \quad \langle \ell_i | \mathcal{E} = \mu_i \langle \ell_i |$$

apply, truncate back to χ , repeat
(left state propagated with $\mathcal{E}^T$)

$$\mu_0 = \frac{\langle L | \mathcal{E} | R \rangle}{\langle L | R \rangle}$$

output: the dominant pair
 $\langle L |, |R \rangle$ and μ_0

random complex seeds; a point is accepted only if
independent
seeds agree and the phase continues smoothly in T



Resolving the Tower: the Block Power Method

propagate k right and k left states; approximate an eigenvector inside the block:

$$|\psi\rangle = \sum_{j=1}^k v_j |R_j\rangle$$

$$\sum_j M_{ij} v_j = \theta \sum_j S_{ij} v_j, \quad S_{ij} = \langle L_i | R_j \rangle, \quad M_{ij} = \langle L_i | \mathcal{E} | R_j \rangle$$

solving the $k \times k$ problem gives the Ritz values

$$\theta_0, \theta_1, \dots, \theta_{k-1} \approx \mu_0, \mu_1, \dots, \mu_{k-1}$$

— the k leading eigenvalues of $\mathcal{E}$, and from their phases the gaps $\Delta\theta_\alpha$

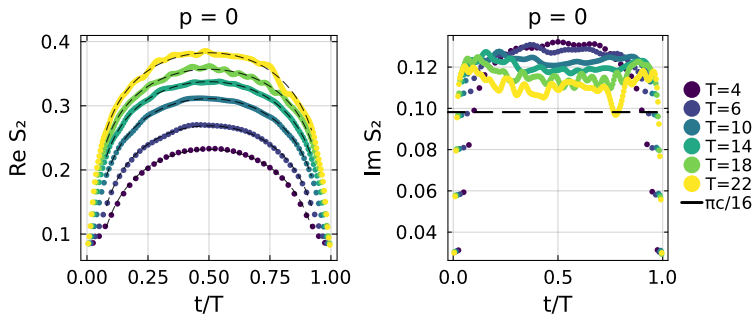
physical branch: modulus window around the previous μ_0 + phase continuity — never largest modulus

Results

the four predictions, measured at every coupling

p	v_∞	entropy window	c (entropy)	spectral window	c (spectral)	x_1
0	2.00	$4 \leq T \leq 22$	0.49 ± 0.01	$2 \leq T \leq 22$	0.45	0.498
0.1	2.67	$4 \leq T \leq 13$	0.50 ± 0.04	$2 \leq T \leq 17$	0.47	0.497
0.3	3.97	$2 \leq T \leq 9$	0.53 ± 0.04	$2 \leq T \leq 10$	0.49	0.493
0.5	5.21	$2 \leq T \leq 5$	0.58 ± 0.04	$2 \leq T \leq 6$	0.53	0.487

Results I — the Entropy Route at $p = 0$



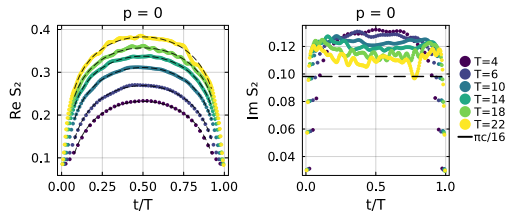
finite- T correction and extrapolation:

$$\text{plateau}(T) = a + b T^{-1/2} \Rightarrow c = \frac{16 a}{\pi}$$

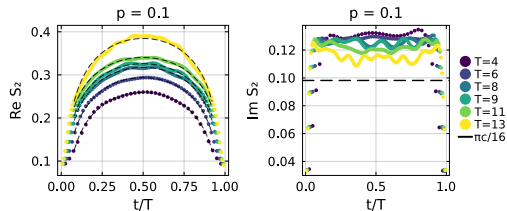
uncertainty: seed scatter of the plateau alone
(resample the runs, refit; 0.01 here, 0.04 at $p > 0$)

p	window	c (entropy)
0	$4 \leq T \leq 22$	0.49 ± 0.01

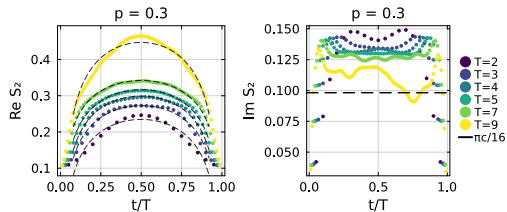
Results I — the Entropy Route at Every Coupling



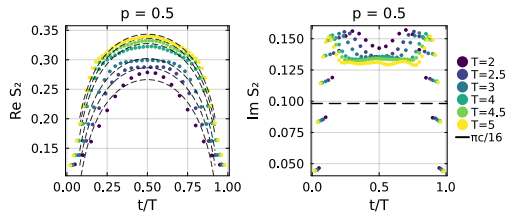
$$c = 0.49 \pm 0.01 \quad (4 \leq T \leq 22)$$



$$c = 0.50 \pm 0.04 \quad (4 \leq T \leq 13)$$



$$c = 0.53 \pm 0.04 \quad (2 \leq T \leq 9)$$



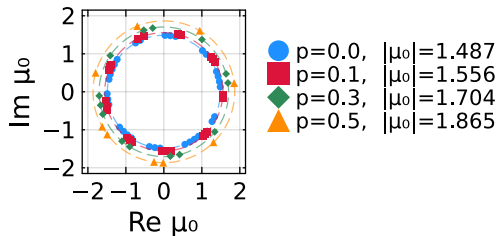
$$c = 0.58 \pm 0.04 \quad (2 \leq T \leq 5)$$

Results II — Emergent Dual Unitarity

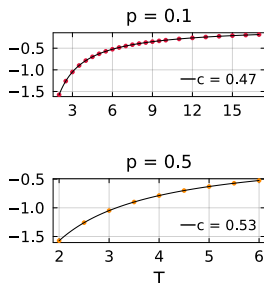
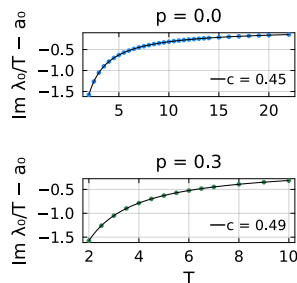
$|\mu_0|$ **constant**
while the phase winds

exactly the predicted coalescence

$$\frac{|\mu_i|}{|\mu_0|} = 1 - \frac{b_i}{T^2} \rightarrow 1$$



Results III — the Central Charge from the Phase



the fit, as implemented:

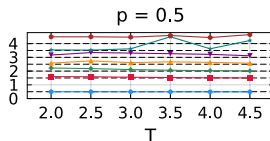
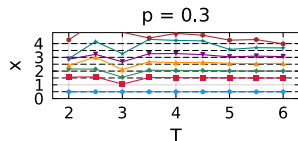
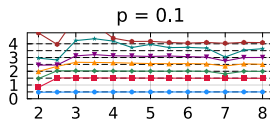
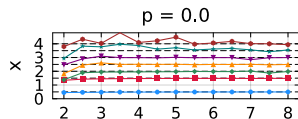
$$\frac{\text{Im } \lambda_0}{T} = a_0 + \frac{B}{T} + \frac{C}{T^2}$$

a_0, C free; $B = -\pi$ fixed (branch);
 $\lambda_0 = \log(-\mu_0)$; $c = 24 \nu |C|/\pi$

p	window	c (spectral)
0	$2 \leq T \leq 22$	0.45
0.1	$2 \leq T \leq 17$	0.47
0.3	$2 \leq T \leq 10$	0.49
0.5	$2 \leq T \leq 6$	0.53

Results IV — the Conformal Boundary Tower

Ising class, free boundaries: $x_\alpha \in \{\frac{1}{2}, \frac{3}{2}, 2, \frac{5}{2}, 3, \frac{7}{2}, 4\}$



$$\Delta\theta_\alpha = \arg \mu_\alpha - \arg \mu_0 = \frac{\pi v x_\alpha}{T}$$

p	x_1
0	0.498
0.1	0.497
0.3	0.493
0.5	0.487

Conclusions

p	v_∞	entropy window	c (entropy)	spectral window	c (spectral)	x_1
0	2.00	$4 \leq T \leq 22$	0.49 ± 0.01	$2 \leq T \leq 22$	0.45	0.498
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Thank you for your attention

Backup A1 — the Entropy Predictions, Derived

$$\text{Tr} \rho^n = \text{replica surface} \Rightarrow \text{twist operators, } x_n = \frac{c}{12} \left(n - \frac{1}{n} \right)$$

cut anchored at the edge $\Rightarrow$ ONE twist + its mirror image $\Rightarrow$ coefficient halved: $\frac{c}{12} \left(1 + \frac{1}{n} \right)$

$$S_n = \frac{c}{12} \left(1 + \frac{1}{n} \right) \log \left[\frac{2\ell}{\pi} \sin \frac{\pi x}{\ell} \right] + s_0 \quad (\log \text{ map: separations} \rightarrow \text{chords})$$

continue $\ell \rightarrow i \nu T$: the ratio $x/\ell = t/T$ stays real, only $\log i = i\pi/2$ is new

$$\text{Re } S_2 = s_0 + \frac{c}{8} W(t, T) \quad (\text{dome}) \quad \text{Im } S_2 = \frac{\pi c}{16} \quad (\text{plateau})$$

Backup A2 — the Spectrum, Derived I: from $T(z)$ to μ_i

$$T(w) = \left(\frac{dz}{dw}\right)^2 T(z) + \frac{c}{12}\{z; w\}, \quad \{z; w\}_{\log} = -\frac{1}{2}\left(\frac{\pi}{\beta}\right)^2$$
$$\implies \hat{H} = \frac{\pi}{\beta}\left(L_0 - \frac{c}{24}\right) \implies \mu_i = \exp\left[-\frac{\pi}{\beta}\left(x_i - \frac{c}{24}\right)\right]$$

on the lattice, two non-universal additions — both the same for every level:

$$E_i = \underbrace{f\beta}_{\text{bulk}} + \underbrace{f_s}_{\text{boundary}} + \frac{\pi}{\beta}\left(x_i - \frac{c}{24}\right), \quad \log \mu_i = -E_i$$

level-independent terms cancel in every difference — differences isolate $x_i - x_0$

Backup A3 — the Spectrum, Derived II: the Expansion We Fit

$$\beta = v(2\beta_0 + iT) \quad \frac{1}{\beta} = -\frac{i}{vT} \sum_{k \geq 0} \left(\frac{2i\beta_0}{T} \right)^k$$

k	goes to	order
0	phase	$1/T$
1	modulus	β_0/T^2
2	phase	β_0^2/T^3

$$\text{Im } \lambda_0 = A T + B + \frac{C}{T} + \mathcal{O}(T^{-3}), \quad A = -fv, \quad B = -\pi, \quad C = -\frac{\pi C}{24v}$$

no $1/T^2$ term (that order is real $\rightarrow$ modulus); B is pure branch (free fit: $B/\pi = -1.0000$)
 the fit **truncates this expansion** $\Rightarrow$ part of the systematic on c

$$\frac{|\mu_i|}{|\mu_0|} = 1 - \frac{2\pi\beta_0(x_i - x_0)}{v T^2} \longrightarrow 1 \quad (\text{the circle})$$

Backup B1 — the Chain in Fermions: Where Integrability Breaks

$$H = - \sum_i \left[\sigma_i^z \sigma_{i+1}^z + \lambda \sigma_i^x + p (\sigma_i^z \sigma_{i+2}^z + \lambda \sigma_i^x \sigma_{i+1}^x) \right]$$

Jordan–Wigner: $\sigma_j^x = 1 - 2n_j$, $\sigma_j^z = (c_j + c_j^\dagger) K_j$, $K_j = \prod_{k < j} (1 - 2n_k)$

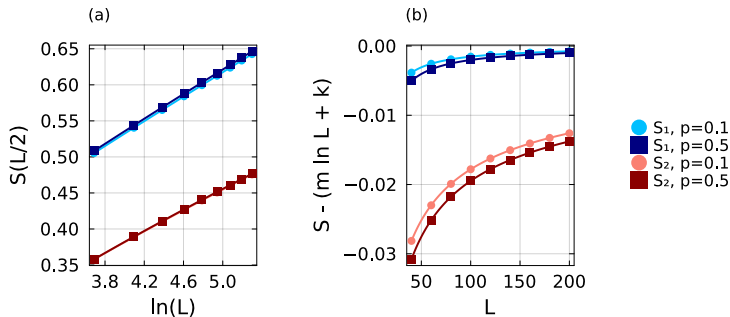
$$H = - \sum_i \left[\underbrace{(c_i^\dagger - c_i)(c_{i+1} + c_{i+1}^\dagger)}_{\text{NN: quadratic}} + \lambda \underbrace{(1 - 2n_i)}_{\text{field}} + p \underbrace{(c_i^\dagger - c_i)(1 - 2n_{i+1})(c_{i+2} + c_{i+2}^\dagger)}_{\text{NNN: contains a quartic piece}} + p\lambda \underbrace{(1 - 2n_i - 2n_{i+1} + 4n_i n_{i+1})}_{\sigma^x \sigma^x: \text{density-density}} \right]$$

the p -terms carry four-fermion pieces $\Rightarrow$ Wick fails — no free-fermion solution for $p > 0$

Backup B2 — Equilibrium c : Finite-Size Scaling

$$S_n(L/2) = \frac{c}{12} \left(1 + \frac{1}{n}\right) \ln L + k_n + A_n L^{-x/n} \quad (\text{mid-chain cut, } x = 1 \text{ fixed})$$

same correction, different decay: L^{-x} for vN, only $L^{-x/2}$ for R2 — why raw R2 overshoots
(0.584) $\Rightarrow$ corrected: $c = 0.500$ (vN), 0.508 (R2)

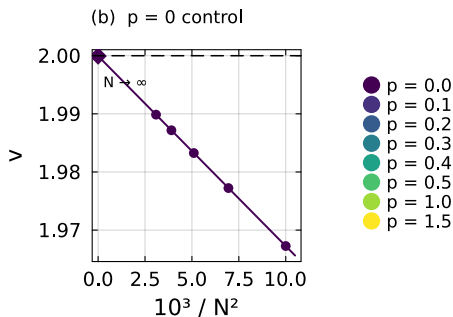
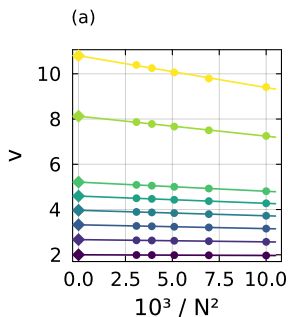


Backup B3 — Sound Velocity: Extrapolation

$$v(N) = \frac{N}{2\pi} [E_1(2\pi/N) - E_1(0)]$$

$$v(N) = v_\infty + \frac{C}{N^2}$$

empirical term: exponent fixed so the extrapolation reproduces $v = 2$ exactly at $p = 0$

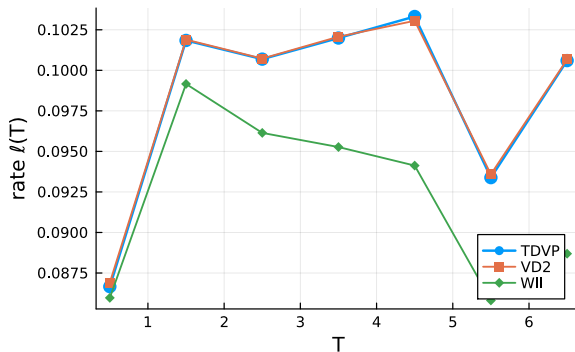


Backup C1 — Schrödinger-Picture Benchmark of the MPO

the Loschmidt rate we compute:

$$\ell(T) = - \lim_{N \rightarrow \infty} \frac{1}{N} \log |\mathcal{L}(T)|$$

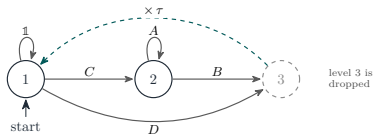
finite chain, $N = 40$, $p = 0.1$:
repeated application of $U(\delta t)$ vs a
TDVP reference



Backup C2 — the Overlap Only the Pair Machine Can Host

the concrete case: $\sigma_3^z \sigma_4^z$ and $\sigma_4^z \sigma_5^z$ share site 4

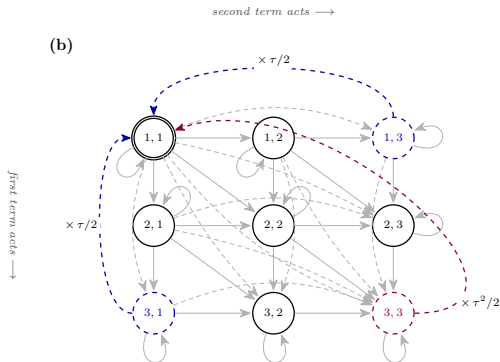
(a)



site 4 would need *B* and *C* — no such move exists:

the overlap is absent at τ^2

(b)



$(1, 1) \xrightarrow{C} (2, 1) \xrightarrow{BC} (3, 2) \xrightarrow{B} (3, 3) \xrightarrow{\text{fold}} \frac{\tau^2}{2}$ (swap the copies = the other ordering)

Backup C3 — Norm Drift Under Repeated Application

$|X+\rangle^{\otimes 100}$, $p = 0.1$, $\delta t = 0.05$, cutoff 10^{-12} after every layer

	W^I	W^{II}	VD2
virtual bond	4	4	13
$\ \psi\ $, 1 layer	1.3514	1.1341	0.99995
$\ \psi\ $, 8 layers	12.6146	3.0181	0.99923
$\ \psi\ $, 16 layers	307.3900	10.8328	0.99751

exact evolution is unitary, so any drift from 1 is pure construction error;
reaching $T = 10$ at this step takes ≈ 200 layers

Backup C4 — Convergence Orders, Measured

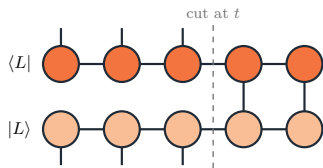
distance to the exactly diagonalised state ($N = 12$, $T = 1$, $p = 0.1$)

δt	W^I		W^{II}		VD2	
	error	order	error	order	error	order
0.2	2.01×10^1	—	2.54	—	1.09×10^{-1}	—
0.1	4.43	2.18	9.29×10^{-1}	1.45	2.44×10^{-2}	2.16
0.05	1.36	1.71	3.93×10^{-1}	1.24	5.93×10^{-3}	2.04
0.025	5.31×10^{-1}	1.35	1.80×10^{-1}	1.12	1.47×10^{-3}	2.01
0.0125	2.36×10^{-1}	1.17	8.65×10^{-2}	1.06	3.68×10^{-4}	2.00
0.00625	1.12×10^{-1}	1.08	4.24×10^{-2}	1.03	9.24×10^{-5}	2.00

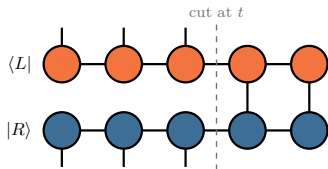
$$\underbrace{\delta t^{q+1}}_{\text{one layer}} \times \underbrace{\frac{T}{\delta t}}_{\text{layers}} = T \delta t^q \implies \frac{\text{err}(2\delta t)}{\text{err}(\delta t)} = 2^q, \quad \log_2 \left[\frac{\text{err}(2\delta t)}{\text{err}(\delta t)} \right] = q$$

(Paeckel et al. 2019); Schrödinger-picture layers, no transverse machinery involved

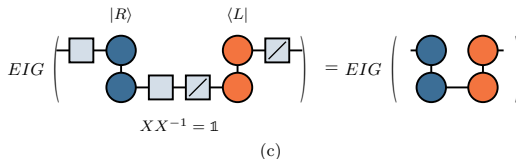
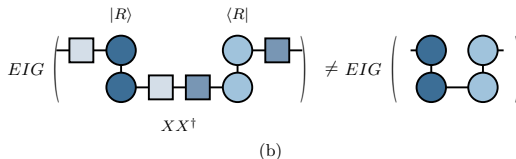
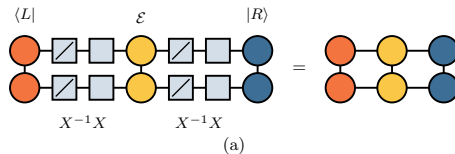
Backup D1 — RDM vs RTM: the Gauge Argument



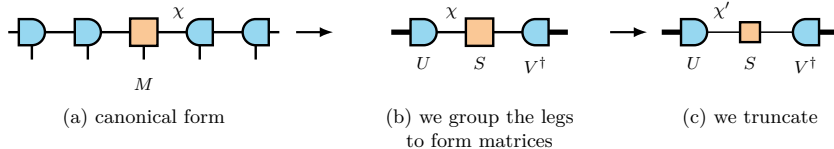
(a) $\rho_L = \text{Tr}_{T-t} |L\rangle \langle L|$



(b) $\mathcal{T}_t = \text{Tr}_{T-t} |R\rangle \langle L| / \langle L|R\rangle$



Backup D2 — RDM Truncation: from the State to a $\chi \times \chi$ Problem



- ▶ mixed canonical form $\Rightarrow$ the cut tensor M ; its SVD $M = USV^\dagger$ is the Schmidt decomposition
- ▶ the surrounding isometries contract to $\mathbb{1}$: $\rho_B = U_B M_A U_B^\dagger$ — the full RDM's spectrum lives in a $\chi \times \chi$ environment
- ▶ eigenvalues = squared singular values, one-to-one $\Rightarrow$ keep the χ' largest (optimal; error = discarded weight)

Backup D3 — RTM Truncation: the Same Reduction, Different Isometries

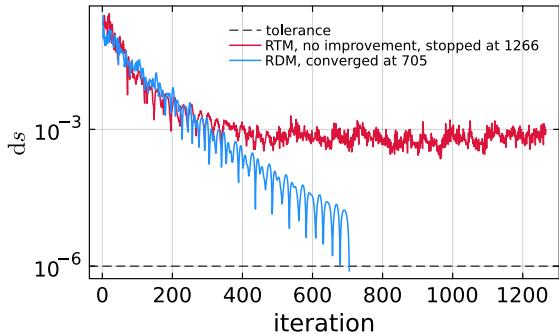
$$\begin{array}{ccc}
 \begin{array}{c} \langle L| \\ |R\rangle \end{array} \begin{array}{c} \text{orange semi-circle} \\ \text{blue semi-circle} \end{array} = \text{C} & \begin{array}{c} SV \left(\begin{array}{c} \text{orange semi-circle} \quad \text{orange semi-circle} \\ \text{blue semi-circle} \quad \text{blue semi-circle} \end{array} \right) = EIG^{1/2} \left(\begin{array}{c} \text{blue semi-circle} \quad \text{blue semi-circle} \\ \text{orange semi-circle} \quad \text{orange semi-circle} \\ \text{orange semi-circle} \quad \text{orange semi-circle} \\ \text{blue semi-circle} \quad \text{blue semi-circle} \end{array} \right) = EIG^{1/2} \left(\begin{array}{c} \text{blue semi-circle} \quad \text{blue semi-circle} \\ \text{orange semi-circle} \quad \text{orange semi-circle} \\ \text{orange semi-circle} \quad \text{orange semi-circle} \\ \text{blue semi-circle} \quad \text{blue semi-circle} \end{array} \right) = SV \left(\begin{array}{c} \text{orange semi-circle} \\ \text{blue semi-circle} \end{array} \right) \end{array} & \begin{array}{c} \text{blue semi-circle} \quad \text{blue semi-circle} \\ \text{orange semi-circle} \quad \text{orange semi-circle} \\ \text{orange semi-circle} \quad \text{orange semi-circle} \\ \text{blue semi-circle} \quad \text{blue semi-circle} \end{array} \right) = EIG^{1/2} \left(\begin{array}{c} \text{blue semi-circle} \quad \text{blue semi-circle} \\ \text{orange semi-circle} \quad \text{orange semi-circle} \\ \text{orange semi-circle} \quad \text{orange semi-circle} \\ \text{blue semi-circle} \quad \text{blue semi-circle} \end{array} \right) = SV \left(\begin{array}{c} \text{orange semi-circle} \\ \text{blue semi-circle} \end{array} \right) \end{array}
 \end{array}$$

(a) (b)

- ▶ generalised canonical form: each $\langle L|$ tensor is paired with the corresponding $|R\rangle$ tensor — the isometries on the two sides differ
- ▶ local reduction: $\tau_B = U_B M_A V_B$ with $U_B \neq V_B^\dagger \Rightarrow$ eigenvalues NOT preserved, but $sv(\tau_B) = sv(M_A)$
- ▶ so we truncate the $\chi \times \chi$ environment on its **singular values**; for the RTM these are no longer one-to-one with the eigenvalues (non-Hermitian) — no variational bound

Backup D4 — How the Iteration Stops

1. ds = change in the singular values of the temporal state between successive iterations — the convergence criterion in the code
2. stop when ds falls below tolerance, or after 150 iterations without improvement
3. RTM hits a noise floor (no-improvement stop) while RDM converges — yet both return the same leading eigenvalue to better than 1 part in 10^4



Backup E1 — the Block Iteration, Step by Step

Algorithm (App. F, Alg. 1): the k dominant eigenpairs of $\mathcal{E}$

1. $|R_j\rangle, \langle L_j| \leftarrow$ random complex tMPS, $j = 1, \dots, k$
2. **apply:** $|\tilde{R}_j\rangle \leftarrow \text{truncate}_\chi(\mathcal{E} |R_j\rangle), \quad \langle \tilde{L}_j| \leftarrow \text{truncate}_\chi(\langle L_j| \mathcal{E})$
3. **project:** $S_{ij} = \langle L_i | R_j \rangle, \quad M_{ij} = \langle L_i | \mathcal{E} | R_j \rangle$
4. solve $Mv = \theta Sv$ (via $\text{pinv}(S)M$) for Ritz values $\theta_1, \dots, \theta_k$ and coefficients V, U
5. **separate:** $|R_j\rangle \leftarrow \text{truncate}_\chi(\sum_i V_{ij} |\tilde{R}_i\rangle)$, analogously $\langle L_j|$ with U
6. normalise; repeat 2–5 until the leading Ritz values stop changing within tolerance

Backup F1 — Why the Spectral c Sits Below $\frac{1}{2}$ at $p = 0$

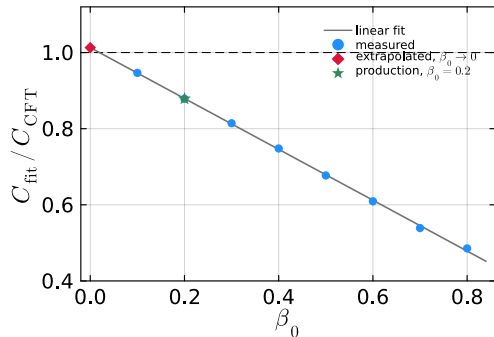
1. the fit truncates the expansion

$$\frac{\text{Im } \lambda_0}{T} = a_0 - \frac{\pi}{T} - \underbrace{\frac{\pi c}{24 v T^2}}_{c \text{ from here}} + \underbrace{\frac{D}{T^4}}_{\text{dropped}} + \dots$$

dropped term = D/T^3 in $\text{Im } \lambda_0$

restoring it: 90% $\rightarrow$ 97% of the
predicted coefficient

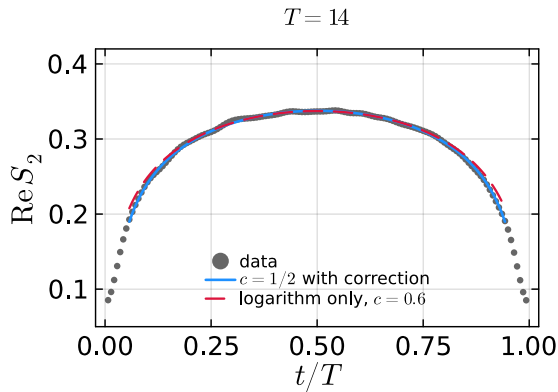
2. the regulator β_0 is finite



linear in β_0 ; $\rightarrow 1.013 \times$ CFT at $\beta_0 \rightarrow 0$

Backup F2 — Why the Real Part Cannot Measure c

$$\operatorname{Re} S_2 = s_0 + \frac{c}{8} W + a w^{-1/2}$$



Backup F3 — Four Boundary Condition Pairs

a **tower** = the set of boundary operators a given pair of boundary conditions allows, and their descendants

